540'145'023 calculated drives - the most used setup tool

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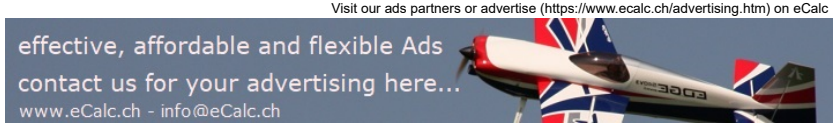

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*prop**Calc** - Propeller Calculator* News (index.htm#news) | Toolbox (index.htm#toolbox) | Easy View (motorcalc_mobile.php) | Help (calcinclude/help/propcalchelp.htm) | Submit Specs (calcmember/submitmotor.htm) | Language: english

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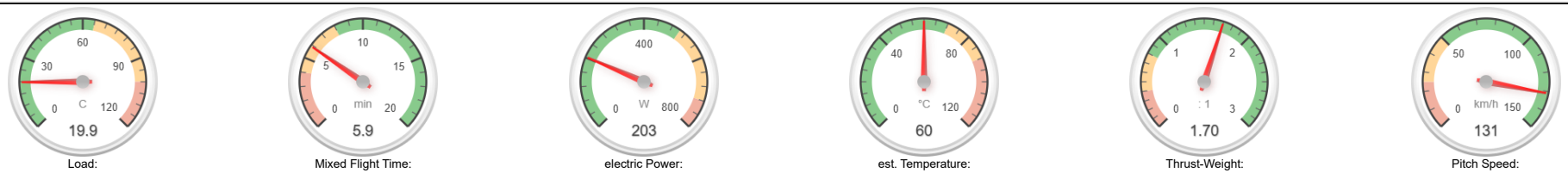
General	Model Weight: 520 g incl. Drive; 18.3 oz	# of Motors: 1 (on same Battery)	Wingspan: 762 mm; 30.00 inch	Wing Area: 17.42 dm²; 270 in²	Drag: coeffcient; 0.18 Cd	Field Elevation: 0 m.ASL; 0 ft.ASL	Air Temperature: 25 °C; 77 °F	Pressure (QNH): 1013 hPa; 29.91 inHg
Battery Cell	Type (Cont. / max. C) - charge state: custom - normal	Configuration: 3 S 1 P	Cell Capacity: 1000 mAh; 1000 mAh total	max. discharge: 85%	Resistance: 0.0113 Ohm	Voltage: 3.7 V	C-Rate: 65 C cont.; 100 C max	Weight: 34 g; 1.2 oz
Controller	Type - Timing: max 20A - normal	Current: 20 A cont.; 20 A max	Resistance: 0.01 Ohm	Weight: 25 g; 0.9 oz	Battery extension Wire: AWG10=5.27mm²	Length: 152.4 mm; 6 inch	Motor extension Wire: AWG12=3.30mm²	Length: 76.2 mm; 3 inch
Motor	Manufacturer - Type (Kv) - Cooling: (² = discontinued) EMAX - EcoII 2207-2400 (2400); medium; search...	KV (w/o torque): 2400 rpm/V; Prop-Kv-Wizard	no-load Current: 2.2 A @ 10 V	Limit (up to 15s): 720 W	Resistance: 0.041 Ohm	Case Length: 20 mm; 0.79 inch	# mag. Poles: 14	Weight: 34 g; 1.2 oz
Propeller	Type - yoke twist: Master Airscrew GF - 0°	Diameter: 5.5 inch; 139.7 mm	Pitch: 4 inch; 101.6 mm	# Blades: 2	Simulation based on: PConst / TConst; 1.06 / 1.0	Gear Ratio: 1 : 1	Flight Speed: 0 km/h; 0 mph	calculate



Load: | Mixed Flight Time: | electric Power: | est. Temperature: | Thrust-Weight: | Pitch Speed:

Remarks:

- The max. current is just below the continuous limit of the controller. In partial load operation, the controller may overheat.
  We suggest excellent ESC cooling or at least 20% headroom for heavy partial load operation (e.g. 3D).
- Excessive wiring extension between battery and ESC leads to high voltage spikes and may harm your ESC input.
  Recommendation: do the ESC supply without wiring extension or install additional low ESR capacitors to the wiring to protect the ESC input.
- The airflow at the propeller blade will stall. Therefore the static thrust and max. current may not be reached. On ground you will measure *Stall Thrust* as maximum.
- 11.0km/h / 6.8mph - above this airspeed stall at the propeller blade will have disappeared completely.

Battery		**Motor @ Optimum Efficiency**		**Motor @ Maximum**		**Propeller**		**Total Drive**		**Airplane**	
Load:	19.87 C	Current:	20.63 A	Current:	19.87 A	Static Thrust:	882 g	Drive Weight:	177 g	All-up Weight:	520 g
Voltage:	10.43 V	Voltage:	10.16 V	Voltage:	10.19 V		31.1 oz		6.2 oz		18.3 oz
Rated Voltage:	11.10 V	Revolutions*:	21381 rpm	Revolutions*:	21544 rpm	Revolutions*:	21544 rpm	Power-Weight:	424 W/kg	Wing Load:	29.9 g/dm²
Energy:	11.1 Wh	electric Power:	209.5 W	electric Power:	202.5 W	Stall Thrust:	692 g		192 W/lb		9.8 oz/ft²
Total Capacity:	1000 mAh	mech. Power:	170.7 W	mech. Power:	164.9 W		24.4 oz	Thrust-Weight:	1.70 : 1	Cubic Wing Load:	7.2
Used Capacity:	850 mAh	Efficiency:	81.5 %	Efficiency:	81.4 %	avail.Thrust @ 0 km/h:	692 g	Current @ max:	19.87 A	est. Stall Speed:	25 km/h
min. Flight Time:	2.6 min			est. Temperature:	60 °C	avail.Thrust @ 0 mph:	24.4 oz	P(in) @ max:	220.6 W		16 mph
Mixed Flight Time:	5.9 min				140 °F	Pitch Speed:	131 km/h	P(out) @ max:	164.9 W	est. Speed (level):	71 km/h
Weight:	102 g			Rm:	57.2 mΩ		81 mph	Efficiency @ max:	74.8 %		44 mph
	3.6 oz			Controller	150808 epm	Tip Speed:	567 km/h	Torque:	0.07 Nm	est. Speed (vertical):	27 km/h
							352 mph		0.05 lbf.ft		17 mph
		Wattmeter readings		**Thermal Propagation**		specific Thrust:	3.42 g/W	Climb Capacity:	1211 m	est. rate of climb:	7.9 m/s
		Current:	19.87 A	2 5 11 17 s			0.12 oz/W		3973 ft	(~85...90°)	1548 ft/min
		Voltage:	10.43 V	39 48 56 59 °C		n100W:	18799 rpm				
		Power:	207.2 W	102 117 132 137 °F		n10N:	23875 rpm				

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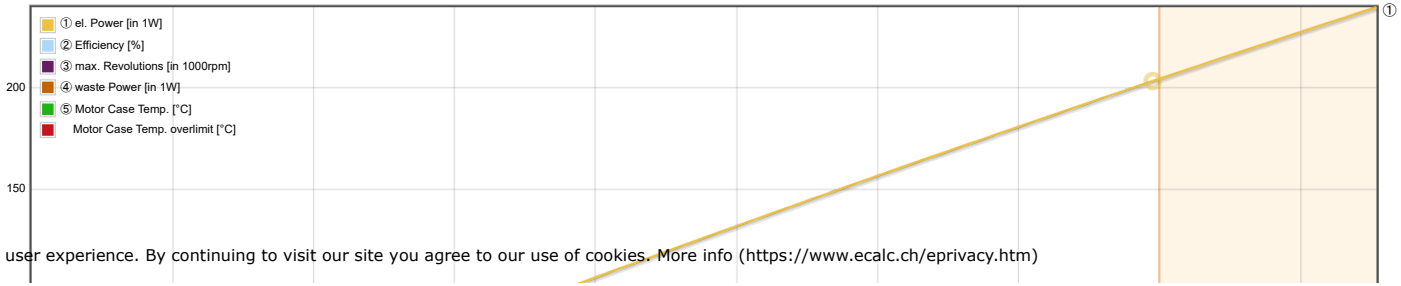
Motor Characteristic at Full Load



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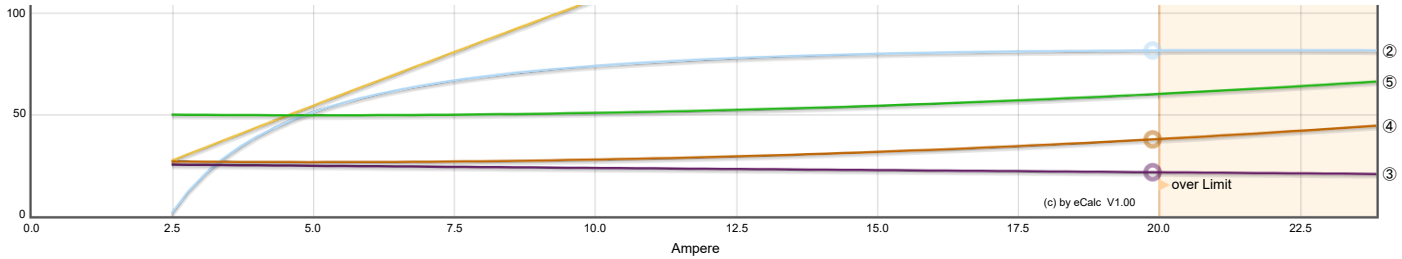
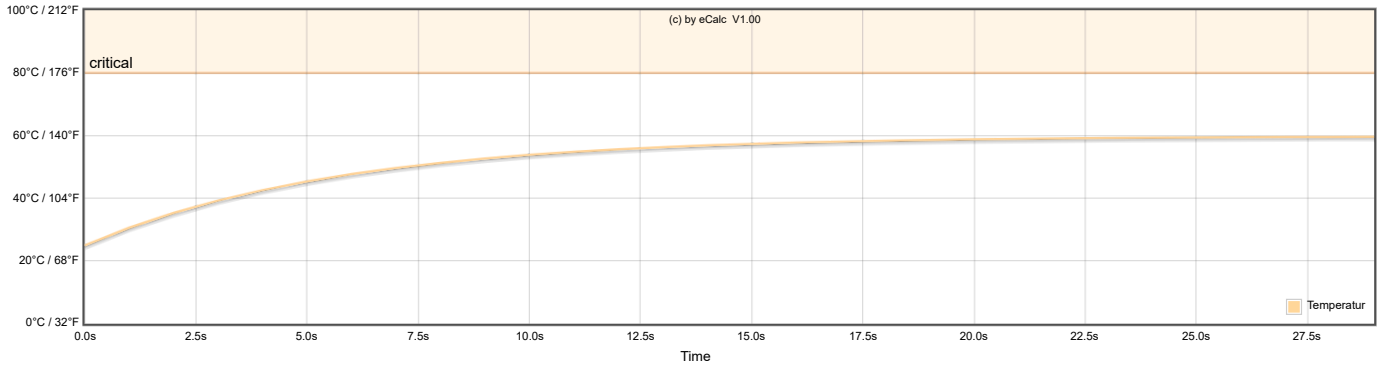


Thermal Propagation at Full Load



Motor Partial Load - Static on Ground

Propeller	Pitch Speed		Throttle	Current (DC)	Voltage (DC)	el. Power	Efficiency	Thrust		Spec. Thrust		Motor Run Time
rpm	km/h	mph	%	A	V	W	%	g	oz	g/W	oz/W	(85%) min
5600	34	21	23	0.5	11.1	5.9	49.1	60	2.1	10.1	0.36	94.7
7000	43	27	29	0.9	11.1	9.6	58.8	93	3.3	9.7	0.34	58.0
8400	51	32	35	1.4	11.1	14.9	65.8	134	4.7	9.0	0.32	37.5
9800	60	37	41	2.0	11.0	22.0	70.7	183	6.4	8.3	0.29	25.3
11200	68	42	47	2.9	11.0	31.2	74.3	238	8.4	7.6	0.27	17.7
12600	77	48	53	4.0	11.0	43.0	76.8	302	10.6	7.0	0.25	12.8
14000	85	53	60	5.4	10.9	57.7	78.6	373	13.1	6.5	0.23	9.5
15400	94	58	67	7.1	10.9	75.5	79.8	451	15.9	6.0	0.21	7.2
16800	102	64	74	9.1	10.8	96.9	80.8	536	18.9	5.5	0.20	5.6
18200	111	69	81	11.6	10.7	122.2	81.4	630	22.2	5.2	0.18	4.4
19600	120	74	89	14.6	10.6	151.8	81.9	730	25.8	4.8	0.17	3.5
21000	**128**	**80**	**97**	**18.1**	**10.5**	**186.1**	**82.1**	**838**	**29.6**	**4.5**	**0.16**	**2.8**
21544	131	82	100	19.9	10.4	202.5	81.4	882	31.1	4.4	0.15	2.6

Motor Partial Load - Dynamic in Flight

Propeller	Speed (level)		Throttle	Current (DC)	Voltage (DC)	el. Power	Efficiency	Thrust		Energy Consumption		Motor Run Time
rpm	km/h	mph	%	A	V	W	%	g	oz	Wh/km	Wh/mi	(85%) min
7833	25	16	31	1.2	11.1	13.0	65.1	91	3.2	0.52	0.84	42.8
9399	30	19	38	1.9	11.0	20.6	71.1	131	4.6	0.69	1.11	26.9
10966	35	22	45	2.9	11.0	31.1	75.0	179	6.3	0.89	1.43	17.8
12532	40	25	52	4.2	11.0	44.8	77.7	234	8.2	1.12	1.80	12.3
14099	45	28	59	5.8	10.9	62.3	79.5	296	10.4	1.38	2.23	8.8
15665	50	31	67	7.9	10.8	84.2	80.6	365	12.9	1.68	2.71	6.4
17232	55	34	75	10.5	10.7	111.0	81.4	442	15.6	2.02	3.25	4.8
18798	60	37	83	13.8	10.6	143.3	81.9	525	18.5	2.39	3.84	3.7
20365	**65**	**40**	**92**	**17.7**	**10.5**	**181.5**	**82.2**	**617**	**21.8**	**2.79**	**4.50**	**2.9**

Important Note:
Before flight recheck your max. current! If your Current, el. Power or RPM are over the manufacturers limits your motor, controller and/or battery may take damage! **Verify before flight by measurment!**

for printing use Landscape format
* The manufacturer limitation is NOT monitored
** Testdata with reduced accuracy

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